

Early Black-Hole Growth Constraints from JWST

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Abstract

Aims. Massive black holes in the early Universe must grow within limited cosmic time. We identify which JWST black-hole candidates most strongly constrain early growth models and which assumptions control these constraints. **Methods.** We compile 350 literature measurement records from 32 source families at $z \geq 4$. We analyse 220 objects with broad hydrogen-line mass estimates and include 14 additional objects in a broader comparison. We propagate reported mass uncertainties and test mass-scale shifts and alternative growth assumptions. **Results.** For $100 M_{\odot}$ seeds formed at $z = 30$, radiative efficiency $\epsilon = 0.1$, and no merger boost, 8 of the 220 objects require a lifetime-average Eddington ratio $\bar{f}_{\text{Edd,req}} > 1$. Of these, 5 exceed this threshold in at least 95% of trials sampling their quoted mass errors under fixed growth assumptions. For UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604, at least 95% of trials still require an average Eddington ratio above 1 after reducing each mass by 0.5 dex (a factor of approximately 3.2). Uncertainties in the mass estimators also affect these constraints. Delaying seeding to $z = 20$ makes GN-z11 the most demanding object. For A2744-QSO1, an independent dynamical mass estimate raises the required average Eddington ratio to approximately 1. At $\epsilon = 0.05719$, the ideal thin-disc efficiency for a non-spinning black hole, all 234 central mass estimates are reachable from $100 M_{\odot}$ seeds formed at $z = 30$ without lifetime-average accretion above the Eddington limit.

Key words. black hole physics – accretion, accretion disks – galaxies: active – galaxies: high-redshift – quasars: supermassive black holes

1 Introduction

Massive black holes observed in the early Universe must have assembled within the available cosmic time. The strength of this constraint depends on the observed black-hole mass and redshift and on assumptions about seed formation, radiative efficiency, and the duration of active accretion.

We compile a literature-based catalogue of JWST-identified high-redshift accreting black holes and candidates at $z \geq 4$, extending from the first billion years to a cosmic age of approximately 1.54 Gyr at its lower-redshift boundary. We retain published measurements, object identity links and mass-estimation sources so that measurements from different studies can be compared.

Analytic seed-growth comparisons already establish the trade-off between seed mass and sustained accretion. For example, [P. Dayal \(2024\)](#) considered light and heavy seeds at $z_{\text{seed}} = 25$ and explored a primordial-seeding interpretation. We apply the same growth equations to a larger catalogue to calculate whether each black hole could reach its observed mass while accreting, on average, at or below the Eddington limit. We test how the required rates change with seed and growth assumptions, masses drawn from published uncertainty intervals, uniform mass shifts, and alternative published measurements. For $100 M_{\odot}$ seeds formed at $z = 30$, radiative efficiency $\epsilon = 0.1$ and no mass added by mergers, 8 objects in our Balmer-line sample require an average Eddington ratio above 1. We list these objects and assess how the limitations reported for their mass estimates affect the growth constraints.

For A2744-QSO1, we compare the catalogue’s virial mass with an independent dynamical estimate.

The contributing studies include the JADES broad-line census (I. Juodžbalis et al. 2026a), which measures current accretion and host scaling relations, and the narrow-line census (J. Scholtz et al. 2025), which identifies candidates for obscured activity. The studies targeted different kinds of objects and reached different detection limits. Their combined probability of selecting and detecting a black hole is unknown. The source-family inventory is given in Appendix A.

1.1 Theoretical framework

We use the luminosity-based Eddington ratio $f_{\text{Edd}} = L_{\text{bol}}/L_{\text{Edd}}$, where L_{bol} is the total luminosity of the accretion flow integrated over all wavelengths and L_{Edd} is the luminosity at which outward radiation force balances gravity for electron scattering in ionized hydrogen. The fixed radiative efficiency ϵ is the fraction of inflowing rest-mass energy emitted as radiation; the fraction $1 - \epsilon$ is retained by the black hole in this model. We define the available growth interval Δt and the lifetime-average Eddington ratio $\bar{f}_{\text{Edd}}$ as

$$\Delta t = t(z_{\text{obs}}) - t(z_{\text{seed}}) > 0, \quad \bar{f}_{\text{Edd}} = \frac{1}{\Delta t} \int_{t(z_{\text{seed}})}^{t(z_{\text{obs}})} f_{\text{Edd}}(t) dt. \quad (1)$$

Here $t(z)$ is the age of the Universe at redshift z , z_{seed} is the redshift at which the seed forms and growth begins, and z_{obs} is the observed redshift. The integration variable t is cosmic time, and $f_{\text{Edd}}(t)$ is the instantaneous Eddington ratio; its average includes both active and quiet phases throughout Δt . We use the exponential growth relation adopted by P. Dayal (2024), expressed in terms of the time-averaged Eddington ratio, and introduce a multiplier B_{merge} to approximate additional mass from mergers:

$$M_{\text{BH}}(t_{\text{obs}}) = B_{\text{merge}} M_{\text{seed}} \exp \left[\frac{1 - \epsilon}{\epsilon} \frac{\bar{f}_{\text{Edd}} \Delta t}{t_{\text{Edd}}} \right], \quad (2)$$

Here $M_{\text{BH}}(t_{\text{obs}})$ is the black-hole mass at the observation time $t_{\text{obs}} = t(z_{\text{obs}})$, and M_{seed} is the initial seed mass. The Eddington timescale, $t_{\text{Edd}} = M_{\text{BH}} c^2 / L_{\text{Edd}} \simeq 0.45$ Gyr, is the time needed to radiate an energy equal to the black hole’s rest-mass energy at its Eddington luminosity, with c denoting the speed of light. Because the Eddington luminosity is proportional to black-hole mass, $t_{\text{Edd}} \simeq 0.45$ Gyr is the same for all black-hole masses under this assumption. The mass-growth e-folding time also depends on efficiency and accretion rate: for constant $f_{\text{Edd}} > 0$, it is $[\epsilon/(1 - \epsilon)] t_{\text{Edd}} / f_{\text{Edd}}$, or 0.05 Gyr for $\epsilon = 0.1$ and $f_{\text{Edd}} = 1$. The dimensionless merger multiplier is $B_{\text{merge}} = 1$ for the reference calculation and 2 for a factor-of-two mass boost. We represent mergers by multiplying the seed mass by a constant factor before calculating accretion. This approximation omits the timing and sequence of mergers, their mass ratios, and mass lost through gravitational radiation. The required lifetime-average ratio $\bar{f}_{\text{Edd,req}}$ is the average needed to reach the observed mass M_{BH} from the assumed seed within Δt :

$$\bar{f}_{\text{Edd,req}} = \max \left[0, \frac{\epsilon}{1 - \epsilon} \frac{t_{\text{Edd}}}{\Delta t} \ln \left(\frac{M_{\text{BH}}}{B_{\text{merge}} M_{\text{seed}}} \right) \right]. \quad (3)$$

If $B_{\text{merge}} M_{\text{seed}}$ exceeds the observed black-hole mass, the logarithm is negative and we set the required average Eddington ratio to zero. Even with zero accretion, this seed and merger contribution predict a mass larger than observed. Reproducing the observed mass would require a smaller seed, a smaller merger contribution, or a model that includes mass loss. This requirement averages accretion over the full growth interval. Inferring a duty cycle also requires assumptions about the rates during active and quiet phases.

We calculate cosmic ages using the present-day Hubble expansion rate $H_0 = 67.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and matter density parameter $\Omega_m = 0.315$, matching these 2 values in P. Dayal (2024), who cite Planck. We impose spatial flatness and set $\Omega_\Lambda = 1 - \Omega_m = 0.685$, where Ω_Λ is the density parameter of the cosmological constant; both density parameters are relative to the present-day critical density. The age calculation includes matter and a cosmological constant and neglects radiation. Appendix B gives the

supporting relations, implementation, and inverse seed-mass diagnostic. Equations 2–3 assume constant radiative efficiency. The ideal thin-disc efficiencies used in Figure 1 are 0.03775, 0.05719, and 0.42265 for dimensionless spins $a = -1, 0, +1$, respectively (J. M. Bardeen et al. 1972). Here a is the black-hole spin parameter: $a = -1$ and $+1$ denote maximal spins counter-aligned and aligned with the disc’s angular momentum, respectively, and $a = 0$ denotes a non-spinning black hole. Each track holds efficiency constant; spin evolution is unspecified.

2 Catalogue construction

We compile 350 literature measurements from 32 source families at $z \geq 4$. Each adopted mass is linked to its source, method, and reported uncertainty. We adopt one measurement per object and retain alternatives for sensitivity tests. Choices follow the documented source assessments. For example, CEERS-2782 uses the higher-signal-to-noise RUBIES-EGS-50052 spectrum preferred by the source study, while NX10835 uses the newly available NEXUS mass estimate. The measurement–object link table records the adopted measurement, publication version, and reason for each choice.

Growth analysis requires a numerical mass and redshift, an identified mass method, and a resolved lensing treatment. Upper limits, indirect model ranges, and masses inferred by assuming an Eddington ratio are excluded from numerical inference. We also exclude 6 catalogue entries in 3 groups where it remains unclear whether entries from different studies refer to the same physical object. Three of these entries have eligible masses. This leaves 220 primary objects and an inclusive exploratory sample of 234 objects.

For the main comparison, we use masses estimated from broad $H\alpha$ or $H\beta$ emission lines in objects with secure or probable evidence for an accreting black hole, using the evidence classifications recorded from the source papers. Seven masses are explicitly flagged as conditional on a broad-line-region interpretation: 6 candidates with intermediate-width $H\alpha$ emission and possible outflow contamination (W. Ren et al. 2025), and RUBIES-EGS-49140 (A. J. Taylor et al. 2025), for which an alternative non-AGN interpretation is recorded. These 7 objects enter the expanded sample only. Four further JADES objects—GS-20057765, GS-20030333, GS-164055, and GN-4685—enter the exploratory sample only because their individual broad- $H\beta$ detections are tentative, although broad emission is supported by the stacked spectrum (I. Juodžbalis et al. 2026a). GLIMPSE-38548 and GLIMPSE-7404 also enter the exploratory sample only; the catalogue classifies them as tentative broad- $H\alpha$ AGN (Q. Fei et al. 2026). After the identity exclusions, the main comparison contains 220 objects, labelled primary in the tables and figures. Other limitations of a virial mass, including possible scattering, are recorded separately. COSMOS3D-13852 remains in the main sample on the basis of its published broad-line AGN identification; we test the reported scattering bias in Section 5.1. These single-epoch virial estimators combine the line width, which estimates gas velocity, with a luminosity-based estimate of the broad-line region’s radius. Their calibration is based on reverberation mapping of nearby AGN (Y. Shen 2013). Using the same line family and mass-estimation framework reduces differences between methods in the comparison. Applying these calibrations at high redshift still requires assessing gas dynamics, geometry, obscuration, and lensing for each object. We also analyse an expanded sample of 234 objects, labelled exploratory in the tables and figures. It includes the 220 primary objects, 13 additional candidates, and GN-z11. GN-z11’s mass is inferred from rest-frame ultraviolet emission lines using local virial relations (R. Maiolino et al. 2024). We include it in the expanded comparison because its lines and calibration differ from those of the Balmer-line sample. The sample division reflects the evidence for accretion and the methods being compared; the reliability of each mass estimate requires an individual assessment. When a paper gives a mass without an uncertainty, we can calculate the average Eddington ratio needed to reach that mass. We cannot calculate the probability that this ratio exceeds 1 from the published information, because no mass uncertainty distribution is available. We therefore include these objects in calculations at their published masses and omit them from probability calculations. The mass estimates retained in each sample still use different calibrations, and

the contributing studies have different target-selection criteria and detection limits. Appendix A gives the source inventory, identity decisions, and sample accounting. Table 1 summarizes the catalogue and analysis sample sizes.

Table 1: Catalogue contents and analysis samples. Measurements are literature records of individual objects; an object may have records from several studies. Object counts are provisional because six entries have uncertain identifications across studies and may represent repeated observations of the same objects.

Category	Count
Measurements / provisional objects / host systems	350 / 338 / 337
Growth-eligible measurements / objects	244 / 237
Primary-method-eligible measurements / objects	234 / 227
Primary sample / inclusive exploratory sample	220 / 234
Objects with mass uncertainties (primary / exploratory)	208 / 222
Objects excluded for uncertain identity	6
Objects without growth-eligible measurements	101

Of the 227 primary-method-eligible objects, 4 tentative JADES detections enter only the exploratory sample and 3 are excluded for uncertain identity, leaving 220 primary objects. The 101 objects without eligible masses include 3 of the 6 identity exclusions. Figure 5 counts those 3 under identity exclusions, leaving 98 in its bar for objects without eligible masses.

3 Growth-model comparison

3.1 Reference model

We calculate the required lifetime-average Eddington ratio using Eq. 3, with $M_{\text{seed}} = 10^2 M_{\odot}$, $z_{\text{seed}} = 30$, $\epsilon = 0.1$, and $B_{\text{merge}} = 1$ as the reference history. We adopt $z_{\text{seed}} = 30$ as an early starting point that allows more time for light-seed growth, and test how later onset changes the required accretion rates. We rank objects by this required average Eddington ratio. Figure 1 illustrates tracks at 4 fixed efficiencies; tracks are selected for proximity to the observed masses. Appendix B.5 specifies the display selection.

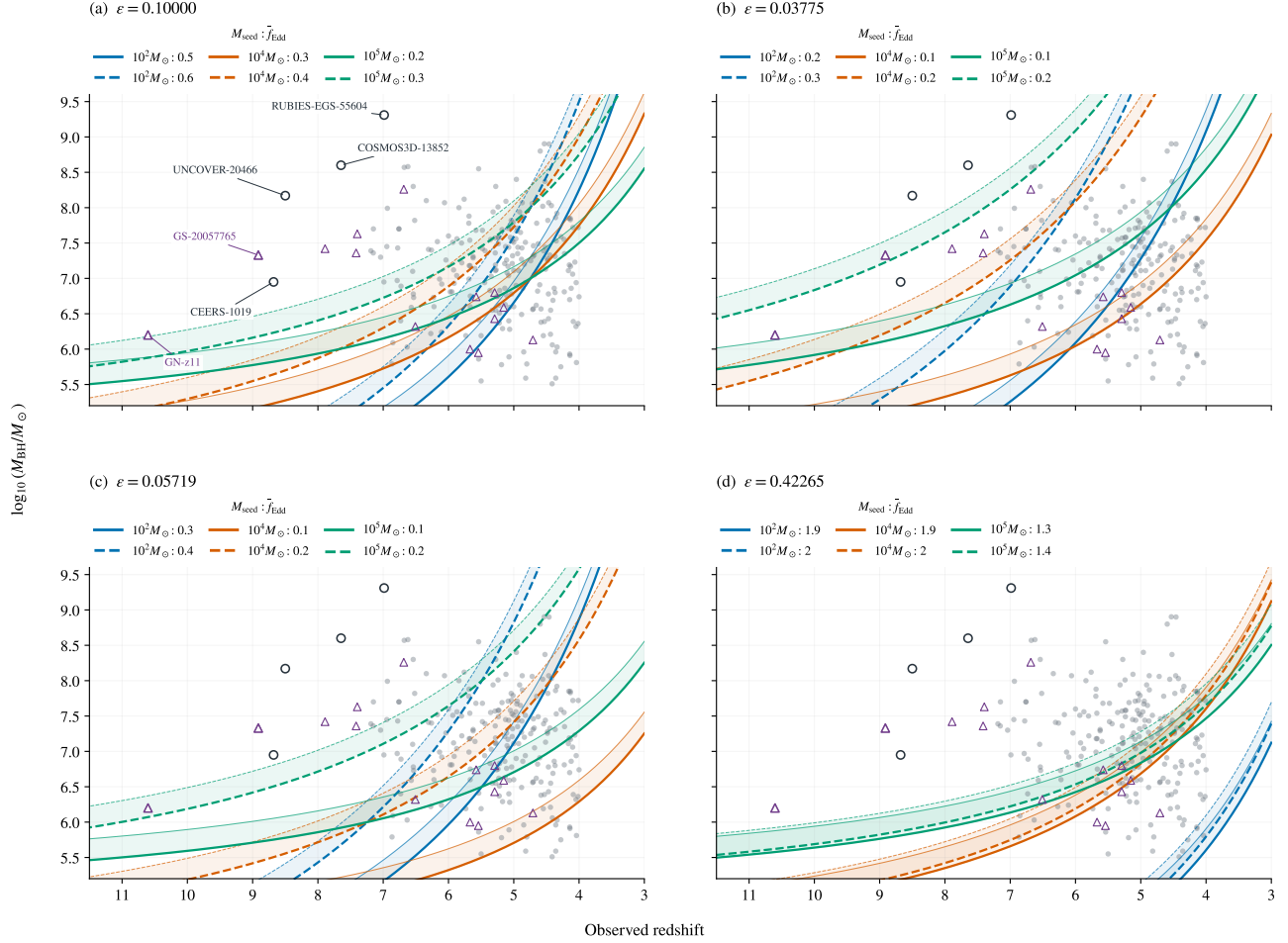
3.2 Uncertainty and sensitivity

We use Monte Carlo sampling to propagate reported asymmetric log-mass errors, with 10,000 draws per object. Each draw has equal probability of falling above or below the central mass, with its distance drawn from a half-normal distribution scaled by the corresponding upper or lower error. Redshifts are held fixed. For each sampled mass, we calculate the average Eddington ratio needed to grow from the assumed seed to that mass. The fraction of the 10,000 Monte Carlo draws with a required ratio above 1 estimates the probability that growth requires an average rate above the Eddington limit, given the adopted mass-error distribution and growth assumptions. Probability calculations use objects with reported mass uncertainties (Table 1).

We test mass-scale sensitivity by shifting all adopted log masses and their error draws by offsets of -0.5 , -0.3 , 0 , $+0.3$, and $+0.5$ dex, keeping the sample and growth assumptions fixed. These fixed offsets are stress tests at the uncertainty scale of single-epoch mass estimates (Y. Shen 2013). For each offset, we report the fraction of shifted Monte Carlo draws that require an average Eddington ratio above 1. Each probability assumes that particular mass shift and the stated seed mass, starting redshift, radiative efficiency, and merger contribution. Sampling and conversion details are given in Appendix B.5.

Growth tracks across radiative-efficiency assumptions

$z_{\text{seed}} = 30$; bands span $B_{\text{merge}} = 1$ (thick edge) to $B_{\text{merge}} = 2$ (thin edge). Seed mass is encoded by colour; exact rates are listed above each panel.



Grey circles: primary (220); purple triangles: exploratory only (14). Outlined targets are named in panel (a).

Figure 1: Growth tracks with 220 primary and 14 exploratory-only objects. Panels show 4 constant radiative efficiencies at $z_{\text{seed}} = 30$; colours distinguish seeds of 10^2 , 10^4 , and $10^5 M_{\odot}$, and legends list the selected average Eddington ratios. Shaded bands span merger boosts $B_{\text{merge}} = 1$ –2, with thick and thin edges respectively. The bands show the effect of the assumed merger boost. Six named objects are outlined in every panel. The reversed redshift axis spans $z = 11.5$ to $z = 3$. Excluded identities and objects without eligible masses are omitted.

4 Results

4.1 Required accretion under the reference assumptions

For a $10^2 M_{\odot}$ seed under the reference history, the median required lifetime-average Eddington ratio across the 220 primary objects is $\bar{f}_{\text{Edd, req}} = 0.573$ (interquartile range 0.485–0.673), with 8 point requirements above unity and none above 2. Across the inclusive 234-object sample, the corresponding median is 0.574 (interquartile range 0.488–0.679), with 14 above unity. These sample statistics use each object’s point requirement. UNCOVER-20466 is most demanding: its point requirement is 1.452, while the median required lifetime-average Eddington ratio over its mass-error draws is 1.453 (16th–84th percentiles 1.354–1.549). Table 2 lists the 8 primary objects above the threshold; their ordering by point requirement differs from ordering by quoted-error probability.

The shorter growth time at $z = 10.603$ places GN-z11 second in the required-rate ranking despite its smaller mass. Its UV virial estimator places it in the exploratory sample.

Table 2: The 8 primary objects above the reference threshold, ordered by their point requirements. Masses and errors are in $\log_{10}(M_{\text{BH}}/M_{\odot})$; all estimators are single-epoch virial. f_0 is the required lifetime-average Eddington ratio evaluated at the adopted central mass under the reference history; f_{16} is the 16th percentile of this required ratio over the object’s mass-error draws, and $P_0 = P(\bar{f}_{\text{Edd,req}} > 1)$. f_- and P_- apply the fixed -0.5 dex offset. Δm (dex) is the reduction in log mass that brings the central required average Eddington ratio to 1, with the growth assumptions unchanged. All probabilities use the quoted mass errors and fixed growth assumptions. Additional mass-estimator limitations are in Table 6.

Object/source	Line	Log mass	f_0	f_{16}	P_0	f_-	P_-	Δm
UNCOVER-20466 J. E. Greene et al. (2024)	H β	$8.17^{+0.42}_{-0.42}$	1.452	1.354	> 0.999	1.335	> 0.999	1.921
COSMOS3D-13852 X. Lin et al. (2026)	H β	$8.60^{+0.35}_{-0.33}$	1.314	1.247	> 0.999	1.214	> 0.999	1.576
RUBIES-EGS-55604 ^a D. D. Kocevski et al. (2025)	H β	$9.31^{+0.10}_{-0.10}$	1.267	1.250	> 0.999	1.180	> 0.999	1.541
CEERS-1019 R. L. Larson et al. (2023)	H β	$6.95^{+0.37}_{-0.37}$	1.205	1.115	0.988	1.083	0.822	0.842
GN-38509 ^b I. Juodžbalis et al. (2026a)	H α	$8.57^{+0.37}_{-0.38}$	1.065	1.005	0.860	0.984	0.392	0.398
J0910_2028_12910 ^a C. J. Baccus & X. Xu (2026)	H α	$8.58^{+0.01}_{-0.01}$	1.053	1.051	> 0.999	0.973	< 0.001	0.330
CEERS-00717 Y. Harikane et al. (2023)	H α	$7.99^{+0.16}_{-0.17}$	1.027	0.998	0.821	0.942	0.018	0.160
ZS7 ^c H. Übler et al. (2024)	H β	$7.70^{+0.40}_{-0.40}$	1.023	0.954	0.625	0.934	0.173	0.130

^a A separate 0.5-dex calibration scale is reported for RUBIES-EGS-55604 and J0910_2028_12910; these scales are omitted from the quoted errors and probabilities shown here. ^b GN-38509’s quoted mass errors already include 0.3-dex calibration scatter. ^c ZS7’s quoted 0.4-dex error already includes calibration scatter. Unmarked rows have no separate numerical calibration scale recorded; their quoted errors may omit other mass-estimator uncertainties.

4.2 Effect of reported mass uncertainties

Of the 8 primary point requirements above unity, 6 remain above it at the 16th percentile and 5 have conditional $P(\bar{f}_{\text{Edd,req}} > 1) \geq 0.95$ (Fig. 2). In the exploratory sample, 14 objects have a median required lifetime-average Eddington ratio above unity, 10 exceed unity at the 16th percentile, and 8 have $P \geq 0.95$; 15 exceed unity at the 84th percentile. Here each median and percentile is taken over the individual object’s mass-error draws. Probabilities follow Section 3.2 and use 208 primary and 222 exploratory objects with reported errors. The 12 primary objects without errors enter only point counts.

Objects above the reference growth threshold

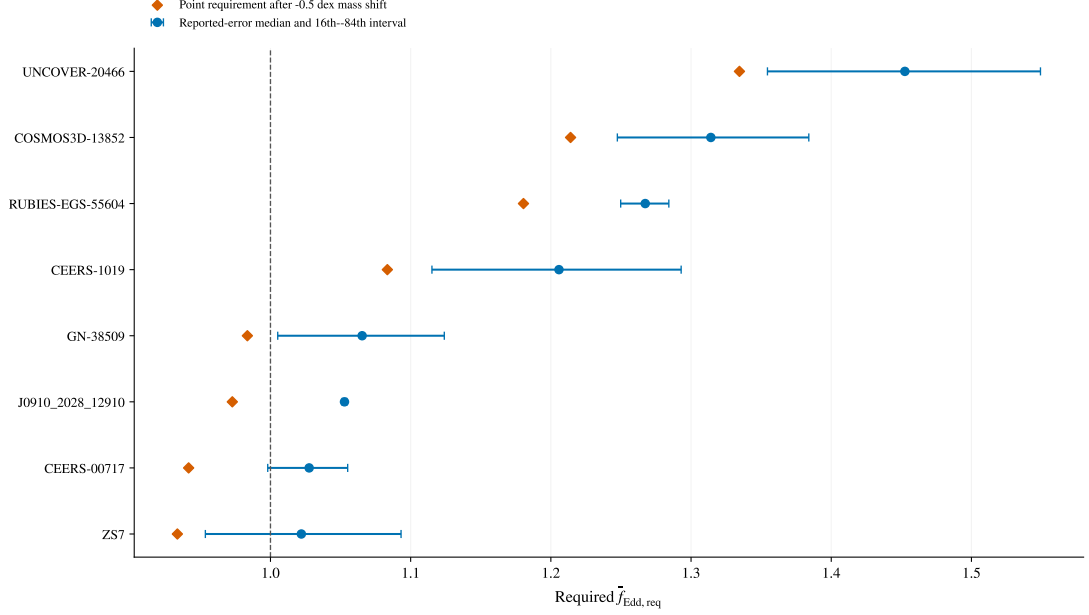


Figure 2: The 8 primary objects whose reference point requirements exceed unity. Circles and bars show medians and 16th–84th percentile intervals of the required lifetime-average Eddington ratio over each object’s mass-error draws; diamonds show point requirements after a -0.5 dex mass shift. The vertical line is the reference threshold. The selected objects all have reported errors; these intervals do not include additional estimator systematics.

4.3 Sensitivity to adopted measurements

Downward mass shifts of 0.3 and 0.5 dex reduce the number of primary objects above unity from 8 to 6 and 4, respectively. The corresponding counts at the 16th percentile fall from 6 to 5 and 3, and the counts with $P \geq 0.95$ fall from 5 to 4 and 3 (Table 3). UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604 retain $P \geq 0.95$ at -0.5 dex. Biases specific to each source may exceed this offset (Section 5.1), and the required rates still depend on seed mass, formation time, and radiative efficiency.

Table 3: Fixed mass-scale stress tests under the reference growth history. The slash-separated counts are for objects above unity at the point estimate, at the 16th percentile, and with conditional $P(\bar{f}_{\text{Edd,req}} > 1) \geq 0.95$. The probability-based counts use 208 primary and 222 exploratory objects with reported errors; point counts use all 220 and 234 objects.

Mass offset (dex)	Primary	Exploratory
-0.5	4 / 3 / 3	7 / 5 / 4
-0.3	6 / 5 / 4	10 / 7 / 5
0	8 / 6 / 5	14 / 10 / 8
$+0.3$	11 / 11 / 10	17 / 15 / 14
$+0.5$	15 / 12 / 11	21 / 18 / 15

We matched 44 catalogue objects to the Baccus & Xu sample using identifiers and sky positions. We replaced our adopted measurements for these objects with the redshifts, black-hole masses, and mass uncertainties reported in Table 1 of C. J. Baccus & X. Xu (2026). Repeating the analysis leaves the primary counts at 8/6/5 and the exploratory counts at 14/10/8, with the same five highest-ranked objects

in the same order in each sample (our Table 4). Of 5 unmatched records, 1 is identity-excluded; retaining or omitting the other 4 leaves these results unchanged (216/230 objects when omitted).

Table 4: Effect of adopting the published Baccus measurements after excluding unresolved identities. Every row excludes all open identity groups. Counts denote point requirements above unity, 16th percentiles above unity, and conditional $P \geq 0.95$, respectively. The adopted catalogue values define the baseline.

Sample	Measurement treatment	Objects	Counts
Primary	Frozen values	220	8/6/5
Primary	Published; retain unmatched	220	8/6/5
Primary	Published; omit unmatched	216	8/6/5
Exploratory	Frozen values	234	14/10/8
Exploratory	Published; retain unmatched	234	14/10/8
Exploratory	Published; omit unmatched	230	14/10/8

4.4 Model compatibility and measurement choice

Increasing the seed mass to $10^3 M_\odot$ reduces the number of primary objects whose required average Eddington ratio exceeds 1 at their adopted central masses from 8 to 3; for $10^5 M_\odot$ seeds, no objects in either sample exceed 1. Delaying seeding to $z = 20$ raises the primary count to 18. Adopting $\epsilon = 0.05719$, the ideal thin-disc efficiency for a non-spinning black hole ($a = 0$; J. M. Bardeen et al. 1972), leaves no objects in either sample requiring an average Eddington ratio above 1 at their adopted central masses (Table 5). Each test changes 1 assumption: seed mass (P. Dayal 2024), available growth time, or radiative efficiency.

Table 5: Objects with point-estimate requirements $\bar{f}_{\text{Edd,req}} > 1$ under fixed histories. Each row changes only the named parameter from the reference ($M_{\text{seed}} = 10^2 M_\odot$, $z_{\text{seed}} = 30$, $\epsilon = 0.1$, $B_{\text{merge}} = 1$). Counts use central mass estimates and provisional object identities.

Scenario	Primary (220)	Exploratory (234)
Reference	8	14
$M_{\text{seed}} = 10^3 M_\odot$	3	5
$M_{\text{seed}} = 10^5 M_\odot$	0	0
$z_{\text{seed}} = 20$	18	24
$\epsilon = 0.05719$	0	0

Three of the 6 records excluded for uncertain identity have eligible masses. Removing these 3 reduces the primary sample from 223 to 220 objects and the exploratory sample from 237 to 234 objects, while preserving threshold counts and top-five rankings in every listed scenario, including the reference probability counts. The median required lifetime-average Eddington ratio across primary objects, evaluated using their central masses, changes from 0.571 to 0.573; the objects above the reference threshold remain the same.

Figure 3 shows the maximum fixed efficiency allowing $\bar{f}_{\text{Edd,req}} \leq 1$ for 5 targets as a function of seed mass. The $z_{\text{seed}} = 20$ and 30 curves show GN-z11’s greater sensitivity to the starting time. These boundaries use central mass estimates and assume no merger boost.

Appendix C gives the fractions of objects whose masses can be reproduced from each seed-mass range under specified accretion rates, spin-dependent efficiencies, and merger contributions.

Seven alternate-measurement comparisons across 6 objects (2 for CEERS-2782) span mass shifts of -0.600 to $+0.949$ dex and shifts in $\bar{f}_{\text{Edd,req}}$ of -0.083 to $+0.121$ (Fig. 4). Both estimates remain below unity in

Seed mass and radiative-efficiency constraints

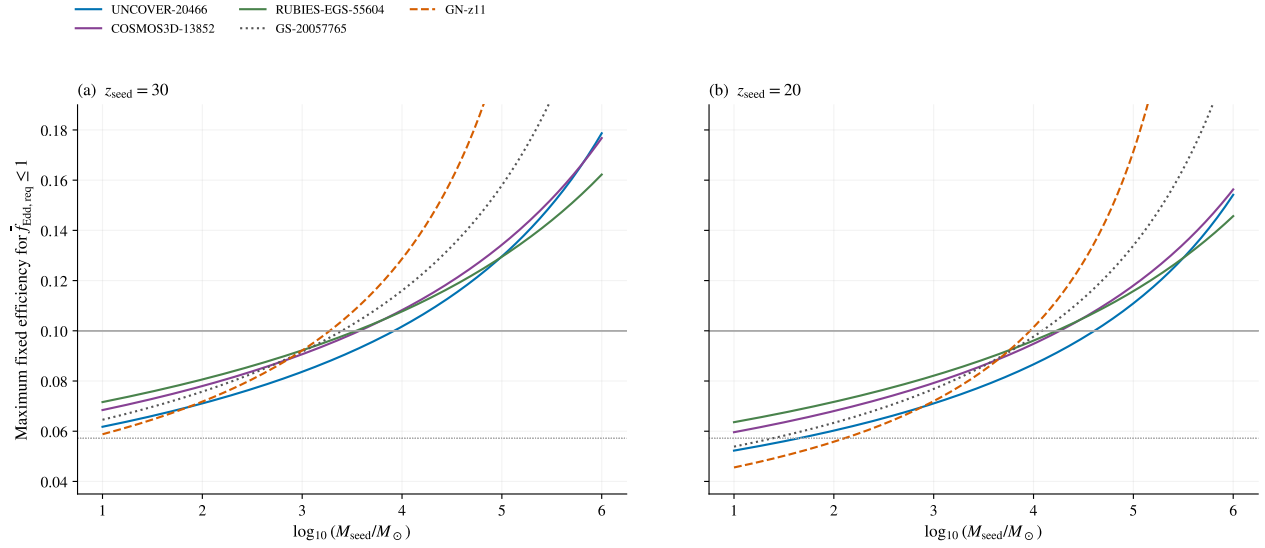


Figure 3: Seed-mass–efficiency boundaries for Eddington-limited average growth at 2 starting redshifts. Each curve uses an adopted point mass and no merger boost. Below a curve the required average is at most unity. Horizontal guides mark $\epsilon = 0.1$ (solid grey) and 0.05719 (dotted grey). GN-z11 (dashed) and GS-20057765 (dotted) are exploratory; the other objects are primary. Each point assumes a radiative efficiency that remains constant throughout growth.

every pair. This comparison covers 6 objects below the threshold. Testing the objects above it requires additional independent mass estimates.

Growth requirements from alternate mass estimates

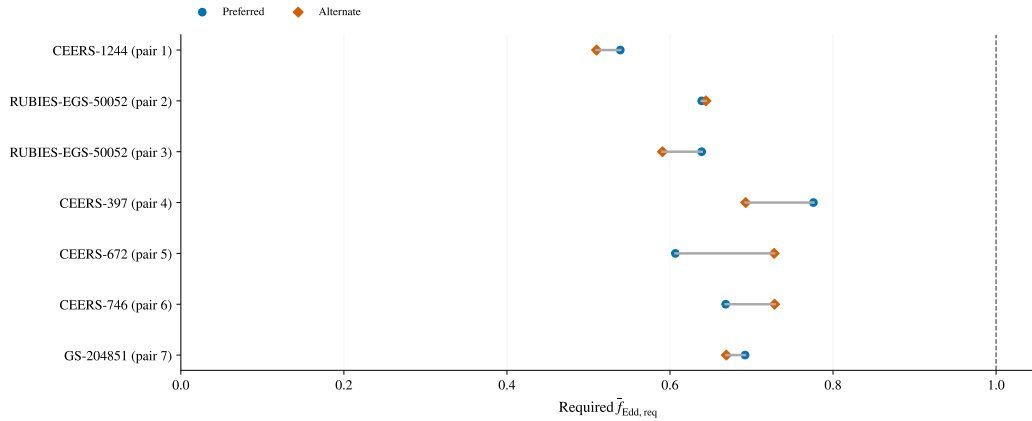


Figure 4: Preferred (circles) and alternate (diamonds) point requirements for 7 measurement pairs across 6 primary objects. Pair numbers follow the released alternate-measurement table, which supplies stable measurement IDs; RUBIES-EGS-50052 (CEERS-2782) has 2 alternatives. All pairs remain below the reference threshold.

4.5 External dynamical-mass check: A2744-QSO1

For A2744-QSO1, we compare the virial mass adopted in the catalogue $\log_{10}(M_{\text{BH}}/M_{\odot}) = 7.3 \pm 0.2$ (J. E. Greene et al. 2024) with the independent MOKA3D dynamical mass estimate 7.7 ± 0.3 (I. Juodžbalis et al. 2026b), using $z = 7.04$ and the reference history. The dynamical estimate depends on lens reconstruction and inclination. We propagate the quoted two-sided mass error; the separate inclination-free lower limit is a one-sided constraint. Using the central mass estimates, the required average Eddington ratio rises from 0.929 to 1.000. With a symmetric normal approximation to the quoted log-mass errors, the exact 16th–84th percentile intervals are 0.895–0.964 and 0.947–1.052, respectively. For the dynamical mass, the unrounded required ratio is just below 1. Its uncertainty interval includes average accretion rates both below and above the Eddington limit. With this mass-error distribution and the growth assumptions held fixed, the probability that the required average Eddington ratio exceeds 1 is 0.497. This comparison tests the mass of 1 catalogue object. Calibrating the virial sample would require independent measurements for more objects.

5 Discussion

These constraints do not uniquely identify a seed-formation history. With radiative efficiency $\epsilon = 0.05719$ for a non-spinning black hole, $100 M_{\odot}$ seeds formed at $z = 30$, and no merger contribution, all adopted central masses can be reached without a lifetime-average Eddington ratio above 1.

5.1 Mass-estimator limitations

Table 6 summarizes the mass-estimator limitations for the 8 primary objects requiring an average Eddington ratio above 1 under the reference assumptions.

Table 6: Mass-estimator limitations for the 8 primary objects requiring $\bar{f}_{\text{Edd,req}} > 1$ under the reference assumptions. Caveats summarize the source records cited in Table 2.

Object	Mass-estimator caveat
UNCOVER-20466	Hbeta; mass depends on lens model and BLR interpretation.
COSMOS3D-13852	Hbeta; source warns of possible 1–2 dex scattering bias.
RUBIES-EGS-55604	Hbeta; dust-corrected mass; separate 0.5 dex calibration scale.
CEERS-1019	Hbeta; broad component has comparatively low significance.
GN-38509	Halpaha; quoted mass errors include 0.3 dex calibration scatter.
J0910_2028_12910	Halpaha; 0.01 dex formal error excludes 0.5 dex mass scale.
CEERS-00717	Halpaha; local calibration; no numeric systematic supplied.
ZS7	Hbeta; quoted 0.4 dex already includes calibration scatter.

Some mass-estimator biases may exceed the 0.5 dex reduction tested above. UNCOVER-20466’s demagnified $\text{H}\beta$ mass depends on lensing and broad-line interpretation (J. E. Greene et al. 2024); RUBIES-EGS-55604’s dust-corrected $\text{H}\beta$ mass has a separately recorded 0.5 dex calibration scale (D. D. Kocevski et al. 2025). For COSMOS3D-13852, scattering could inflate the $\text{H}\beta$ mass by 1–2 dex (X. Lin et al. 2026). Reducing its adopted mass by factors of 10 and 100 lowers its required average Eddington ratio to 1.115 and 0.916, respectively, under the reference assumptions. The size of the correction for this object is uncertain.

Table 2 gives the reduction in each adopted log mass needed to bring its required average Eddington ratio to 1 under the reference assumptions. For J0910_2028_12910, a reduction of 0.330 dex is sufficient. This

is smaller than its separately reported 0.5-dex calibration scale (C. J. Baccus & X. Xu 2026), although its 0.01-dex quoted error gives $P > 0.999$ in the reported-error calculation. COSMOS3D-13852 requires a reduction of 1.576 dex, within the 1–2 dex range of possible scattering bias. UNCOVER-20466 requires 1.921 dex; its lensing and broad-line assumptions still require independent tests. ZS7’s quoted 0.4 dex already includes calibration scatter (H. Übler et al. 2024). The probabilities therefore reflect different combinations of measurement and calibration uncertainty across the sample.

Confirming the weak broad $H\beta$ component in CEERS-1019 would strengthen the basis of its virial mass (R. L. Larson et al. 2023), although calibration uncertainty would remain. An independent mass estimate for J0910_2028_12910 could test whether it still requires an average Eddington ratio above 1 under the reference assumptions. Shared calibration errors would limit how strongly agreement between estimators supports the inferred mass.

5.2 Growth constraints for individual objects

UNCOVER-20466 and GS-20057765 require average Eddington ratios above 1 for $10^3 M_\odot$ seeds, but below 1 for $10^5 M_\odot$ seeds, with the other reference assumptions unchanged. At radiative efficiency $\epsilon = 0.05719$, $100 M_\odot$ seeds formed at $z = 30$ without merger growth require average Eddington ratios of 0.793 and 0.740, respectively. If accretion occurs at the Eddington limit during active phases and ceases between them, these values require activity for about 79% and 74% of the available time. The constraints for GS-20057765 depend on its tentative broad- $H\beta$ component tracing a broad-line region.

GN-z11 requires an average Eddington ratio of 1.437 under the reference assumptions. Delaying seeding to $z = 20$ raises this to 1.880, above UNCOVER-20466’s 1.733, reversing their order. A mass estimate based on the Balmer lines would test whether this result depends on GN-z11’s UV mass estimator. Appendix D shows how seed mass and starting redshift affect the required average Eddington ratios for both objects.

5.3 Limitations

The contributing studies use different selection criteria and detection limits. Inferring population demographics would require a combined completeness model. Virial calibration and geometry errors are incompletely reported. The mass shifts test how the results respond to systematic changes in mass.

Two duplicate pairs have been reconciled using the source papers; the identities of 3 groups remain unresolved (Appendix A.1). Excluding all 6 affected records preserves the threshold results. Reinstating these records requires checking their spectra, masses, images, or apertures against the source papers. Counts of unique objects and hosts remain provisional.

Equation 2 describes growth through an average accretion rate, a fixed merger multiplier, and constant radiative efficiency. Feedback and individual accretion histories remain unspecified. The probabilities hold these assumptions fixed. Averaging over them would require justified parameter distributions and correlations; identifying seed origins would also require formation models and population information.

6 Conclusions

For the adopted masses and growth assumptions, we find:

- For $100 M_\odot$ seeds formed at $z = 30$, $\epsilon = 0.1$, and no merger contribution, 8 of the 220 objects in the main sample require an average Eddington ratio above 1. Five do so in at least 95% of Monte Carlo draws from their quoted mass-error distributions.

- For UNCOVER-20466, COSMOS3D-13852, and RUBIES-EGS-55604, at least 95% of draws still require an average Eddington ratio above 1 after reducing the masses by 0.5 dex. Their mass estimates require tests of lensing, scattering, and calibration, respectively.
- The assumed start of growth changes which objects require average accretion above the Eddington limit and which requires the highest rate. At constant efficiency $\epsilon = 0.05719$, every central mass in the 234-object expanded sample is reachable from a $100 M_{\odot}$ seed at $z = 30$ with an average Eddington ratio below 1.
- For A2744-QSO1, replacing the adopted virial mass with the independent dynamical estimate raises the required average Eddington ratio from 0.929 to approximately 1. Its mass uncertainty allows average rates both below and above the Eddington limit.

Appendix A Source-family inventory

Figure 5 summarizes the sample accounting; Table 7 lists the contributing source families.

Manuscript sample accounting

338 provisional catalogue objects

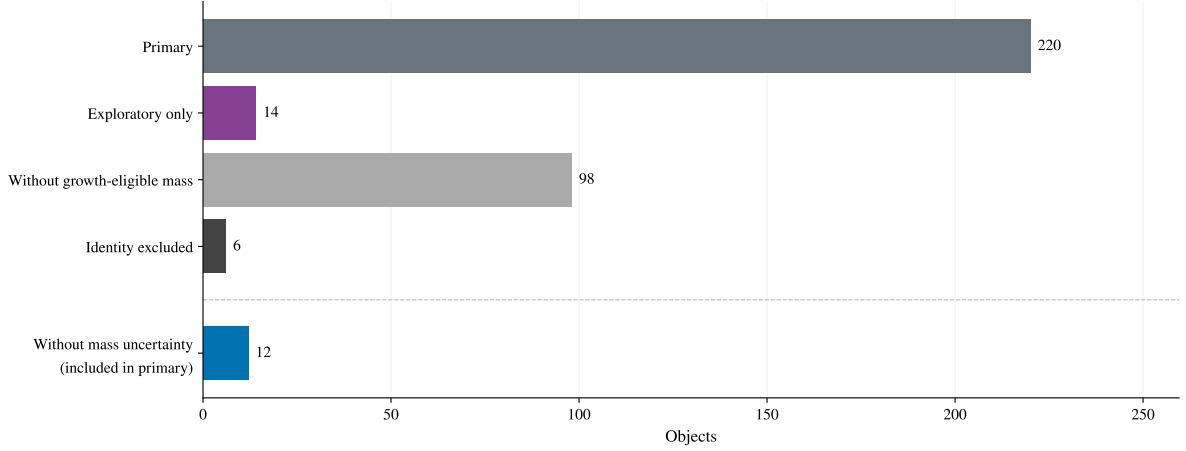


Figure 5: Sample accounting for 338 provisional catalogue objects. The first four bars show mutually exclusive categories: 220 primary objects, 14 additional exploratory objects, 98 retained without eligible masses, and 6 identity exclusions. The final bar shows the 12 primary objects without reported mass uncertainties; these are included in the primary count. The inclusive exploratory sample contains 234 objects. Probability calculations use 208 primary and 222 exploratory objects.

Auxiliary coordinates are attributed in the registry to THRILS (T. A. Hutchison et al. 2025), UHZ1 spectroscopy (A. D. Goulding et al. 2023), and the versioned JADES DR3 GOODS-S prism catalogue.

Table 7: Source-family inventory. Counts are admitted measurements / eligible mass measurements / preferred manuscript primary objects (M/E/P). The columns sum to 350/244/220; E includes alternatives, whereas P counts each selected object once. “Balmer” denotes single-epoch virial masses. “Context” denotes mass estimates retained for discussion but excluded from growth calculations because of their method or interpretation. “Upper limit” denotes an upper bound rather than a measured central mass; these bounds are also excluded from growth calculations.

Source	M/E/P	Mass	Selection channel; principal caveat
C. J. Baccus & X. Xu (2026)	49/49/48	Balmer	NIRSpec broad lines; Frozen preprint masses; calibration
Ó. A. Chávez Ortiz et al. (2026)	1/0/0	Context	UV/optical diagnostics; Model-dependent mass
J. Chisholm et al. (2024)	1/0/0	None	High-ionization lines; No canonical mass
K. Davis et al. (2026) / THRILS	6/6/6	Balmer	EELG-selected spectroscopy; Virial calibration
Q. Fei et al. (2026) / GLIMPSE	10/10/8	Balmer	Lensed broad lines; Lensing corrections
J. E. Greene et al. (2024) / UNCOVER	9/9/9	Balmer	Lensed broad lines; Lens model; BLR interpretation
Y. Harikane et al. (2023)	10/10/5	Balmer	NIRSpec broad lines; Local virial calibration

Table 7 continued

Source	M/E/P	Mass	Selection channel; principal caveat
I. Juodžbalis et al. (2026a) / JADES	23/23/17	Balmer	Broad Balmer lines; Extinction; calibration
M. Killi et al. (2024)	1/1/1	Balmer	Broad Halpha; Linear mass-error conversion
D. D. Kocevski et al. (2025) / LRDs	6/6/6	Balmer	Broad Balmer lines; Dust correction; calibration
R. L. Larson et al. (2023) / CEERS-1019	1/1/1	Balmer	Broad Hbeta; Broad-component significance
X. Lin et al. (2024) / ASPIRE	16/16/16	Balmer	Broad Halpha; Virial calibration
X. Lin et al. (2026) / COSMOS-3D	13/13/13	Balmer	NIRCam grism broad lines; Possible scattering bias
J. Lyu et al. (2024) / SMILES	19/0/0	None	MIRI SED selection; SED-dependent AGN identification
R. Maiolino et al. (2024) / GN-z11	1/1/0	UV	UV lines; UV estimator outside primary
S. Mascia et al. (2026)	8/0/0	None	Compact blue broad emitters; No canonical mass
J. Matthee et al. (2024) / EIGER-FRESCO	20/20/19	Balmer	Broad Halpha; Dust; virial calibration
G. Mazzolari et al. (2025)	25/0/0	None	Narrow-line diagnostics; Star-formation overlap
G. C. K. Leung et al. (2026) / MEOW	12/0/0	None	MIRI SED selection; SED-dependent AGN fraction
R. P. Naidu et al. (2026) / MoM-BH*-1	1/0/0	Context	Broad Hbeta / SED; Scattering; stellar alternative
L. Napolitano et al. (2025a) / GHZ9	1/0/0	Context	UV lines / X-ray evidence; Mass assumes Eddington rate
L. Napolitano et al. (2025b) / GHZ4,7	2/0/0	None	Narrow / high-ionization lines; Tentative diagnostics
W. Ren et al. (2025) / ALPINE-CRISTAL	7/7/1	Balmer	Host-selected broad Halpha; Conditional BLR interpretations
J. Scholtz et al. (2025) / JADES	21/0/0	None	Narrow / high-ionization lines; Tentative classifications
D. D. Kocevski et al. (2026) / Skyfire (CEERS)	22/22/22	Balmer	Broad Halpha; Virial calibration
M. Tang et al. (2025)	3/0/0	None	High-ionization lines; Ionizing-source ambiguity
A. J. Taylor et al. (2025) / CEERS-RUBIES	37/37/35	Balmer	Broad Halpha; Slit loss; dust; calibration
H. Treiber et al. (2025) / UNCOVER	2/0/0	None	UV-line diagnostics; Stellar alternatives
H. Übler et al. (2024) / ZS7	1/1/1	Balmer	Offset broad Hbeta; Error includes calibration scatter
Á. Bogdán et al. (2024); F. Zou et al. (2026) / UHZ1	2/0/0	Context	X-rays / SED; Later detection disputed
Z. Zhang et al. (2026) / LRDs	5/0/0	Upper limit	Narrow lines / SED; Mass upper limits
M.-Y. Zhuang et al. (2026) / NEXUS	15/12/12	Balmer	NIRCam WFSS broad lines; Twelve masses lack errors

A.1 Adopted measurements and uncertain object matches

The catalogue records the source paper, version, quoted uncertainty, and reason for choosing each adopted measurement. For three pairs of entries, it remains unclear whether both refer to the same object: Baccus GDS_1210_9515 and JADES GS-8083; GS-10013704 and Scholtz 99671; and Scholtz 16745 and 208643.

All six entries are excluded from the analysis. Three have usable masses and otherwise meet the primary-sample criteria; the other three have no usable masses. Before including them, their spectra, images, and extraction apertures must be checked to establish which objects were measured. The catalogue documents these decisions and subsequent revisions.

Appendix B Growth-model derivations and implementation

B.1 Accretion and exponential growth

For electron scattering in ionized hydrogen, the Eddington luminosity, Eddington ratio, and Eddington timescale are

$$L_{\text{Edd}} = \frac{4\pi G M_{\text{BH}} m_p c}{\sigma_T}, \quad f_{\text{Edd}} = \frac{L_{\text{bol}}}{L_{\text{Edd}}}, \quad t_{\text{Edd}} = \frac{M_{\text{BH}} c^2}{L_{\text{Edd}}} = \frac{c \sigma_T}{4\pi G m_p} \simeq 0.45 \text{ Gyr}. \quad (4)$$

Here G , c , m_p , and σ_T denote the gravitational constant, speed of light, proton mass, and Thomson cross section. With radiative efficiency $0 < \epsilon < 1$ and rest-mass inflow rate $\dot{M}_{\text{in}}$,

$$L_{\text{bol}} = \epsilon \dot{M}_{\text{in}} c^2, \quad \dot{M}_{\text{BH}} = (1 - \epsilon) \dot{M}_{\text{in}} = \frac{1 - \epsilon}{\epsilon} \frac{f_{\text{Edd}}}{t_{\text{Edd}}} M_{\text{BH}}. \quad (5)$$

The inflow rate is defined at the radiating flow; earlier mass loss is omitted. Dividing Eq. 5 by M_{BH} and integrating at fixed ϵ gives

$$\ln \left(\frac{M_{\text{BH}}}{B_{\text{merge}} M_{\text{seed}}} \right) = \frac{1 - \epsilon}{\epsilon} \frac{1}{t_{\text{Edd}}} \int_{t(z_{\text{seed}})}^{t(z_{\text{obs}})} f_{\text{Edd}}(t) dt = \frac{1 - \epsilon}{\epsilon} \frac{\bar{f}_{\text{Edd}} \Delta t}{t_{\text{Edd}}}. \quad (6)$$

Exponentiating gives Eq. 2; solving for the average rate, with a floor of zero, gives Eq. 3. For a constant positive f_{Edd} , the e-folding time is

$$t_{\text{efold}} = \frac{\epsilon}{1 - \epsilon} \frac{t_{\text{Edd}}}{f_{\text{Edd}}}. \quad (7)$$

It is 0.05 Gyr at $\epsilon = 0.1$ and $f_{\text{Edd}} = 1$.

B.2 Merger multiplier and inverse seed mass

The implementation evaluates Eq. 2 in log mass:

$$\log_{10} \left(\frac{M_{\text{BH}}}{M_{\odot}} \right) = \log_{10} \left(\frac{M_{\text{seed}}}{M_{\odot}} \right) + \log_{10} B_{\text{merge}} + \frac{1 - \epsilon}{\epsilon} \frac{\bar{f}_{\text{Edd}} \Delta t}{t_{\text{Edd}} \ln 10}. \quad (8)$$

We evaluate $B_{\text{merge}} = 1$ (no boost) and 2 (a factor-of-two mass boost). At fixed accretion history,

$$\frac{M_{\text{BH}}(B_{\text{merge}} = 2)}{M_{\text{BH}}(B_{\text{merge}} = 1)} = 2, \quad \Delta \log_{10} M_{\text{BH}} = \log_{10} 2 \simeq 0.3010. \quad (9)$$

Solving the same relation for seed mass gives

$$\log_{10} \left(\frac{M_{\text{seed,req}}}{M_{\odot}} \right) = \log_{10} \left(\frac{M_{\text{BH}}}{M_{\odot}} \right) - \log_{10} B_{\text{merge}} - \frac{1 - \epsilon}{\epsilon} \frac{\bar{f}_{\text{Edd}} \Delta t}{t_{\text{Edd}} \ln 10}. \quad (10)$$

Here $M_{\text{seed,req}}$ is the seed mass required for the specified average Eddington ratio, radiative efficiency, growth interval, and merger multiplier.

B.3 Cosmic time

Cosmic ages use the flat matter-plus-Lambda relation

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}} \operatorname{asinh} \left[\sqrt{\frac{\Omega_\Lambda}{\Omega_m}} (1+z)^{-3/2} \right], \quad (11)$$

with $H_0 = 67.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$. The Hubble constant is converted to inverse time; ages and growth intervals are in Gyr. Radiation is neglected.

B.4 Two-state duty cycle

For a two-state history with the same efficiency in both states,

$$\bar{f}_{\text{Edd}} = Df_{\text{burst}} + (1-D)f_{\text{quiet}}, \quad D_{\text{req}} = \frac{\bar{f}_{\text{Edd,req}} - f_{\text{quiet}}}{f_{\text{burst}} - f_{\text{quiet}}}. \quad (12)$$

Here D is the fraction of time spent active, with $f_{\text{burst}} > f_{\text{quiet}} \geq 0$. The expression applies when $\bar{f}_{\text{Edd,req}} \geq f_{\text{quiet}}$. Values $D_{\text{req}} > 1$ identify histories that cannot reach the observed mass at the assumed rates.

B.5 Mass uncertainties and growth-track plotting

Sampling uses random seed 20260808 and a stable random stream for each object identifier. With 10,000 draws, the Monte Carlo standard error of a probability is at most 0.005, or about 0.0022 at a probability of 0.95; this reflects the finite number of draws. The 12 NEXUS objects without reported mass errors have repeated central values and zero-width stored intervals, but no probabilities or uncertainty ranks. Table 2 distinguishes separate calibration scales from scatter already included in the quoted errors.

The mass-scale tests add the same fixed offset to every original mass draw. The final published Killi mass $8_{-0.4}^{+0.5} \times 10^8 M_\odot$ is converted by transforming both linear bounds, giving $\log_{10}(M_{\text{BH}}/M_\odot) = 8.90309_{-0.02228}^{+0.02633}$.

For Figure 1, we test average Eddington ratios from 0.1 to 2.0 in steps of 0.1. At each seed mass and fixed radiative efficiency, we plot the 2 rates with the most primary objects within 0.5 dex of either merger prediction ($B_{\text{merge}} = 1$ or 2), requiring at least 5 objects. Ties favour smaller median absolute residuals, then smaller rates. Most selected objects lie at $z \simeq 4$ –6. This selection is for display; conclusions use the required-rate ranking. Reference rankings and seed-mass-redshift maps use $\epsilon = 0.1$.

Appendix C Compatibility across seed masses and accretion rates

C.1 Efficiency prescription

The maps use the ideal Kerr thin-disc radiative efficiency at the innermost stable circular orbit (ISCO; J. M. Bardeen et al. 1972):

$$\epsilon_a = 1 - \sqrt{1 - \frac{2}{3r_{\text{ISCO}}}}, \quad (13)$$

$$r_{\text{ISCO}} = 3 + Z_2 - \operatorname{sgn}(a)\sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}, \quad (14)$$

$$Z_1 = 1 + (1 - a^2)^{1/3}[(1 + a)^{1/3} + (1 - a)^{1/3}], \quad Z_2 = \sqrt{3a^2 + Z_1^2}. \quad (15)$$

Here $a = cJ/(GM_{\text{BH}}^2)$ is the dimensionless spin, J is the signed angular momentum relative to the disc, and r_{ISCO} is measured in units of GM_{BH}/c^2 . Neglecting photon capture and stress at the ISCO gives $\epsilon_a = 0.03775$, 0.05719 , and 0.42265 for $a = -1, 0, +1$. For supercritical inflow, we adopt a logarithmic luminosity relation motivated by advection and wind models (J. Poutanen et al. 2007), with coefficient 1 and no radial mass loss:

$$f_{\text{Edd}} = 1 + \ln \dot{m}, \quad \dot{m} = \frac{\dot{M}_{\text{in}}}{L_{\text{Edd}}/(\epsilon_a c^2)}, \quad \epsilon_{\text{eff}} = \begin{cases} \epsilon_a, & f_{\text{Edd}} \leq 1, \\ \epsilon_a f_{\text{Edd}} \exp(1 - f_{\text{Edd}}), & f_{\text{Edd}} > 1. \end{cases} \quad (16)$$

The logarithmic relation applies only for $f_{\text{Edd}} > 1$. Each cell in Figure 6 holds spin, Eddington ratio, and effective radiative efficiency constant, so $f_{\text{Edd}} = \bar{f}_{\text{Edd}}$; relativistic slim-disc dynamics are omitted. If efficiency varies, growth depends on $\int [(1 - \epsilon(t))/\epsilon(t)] f_{\text{Edd}}(t) dt$ and requires both histories.

C.2 Seed-mass ranges and compatible fractions

We solve Eq. 10 for each object’s required seed mass and check whether it lies within each of four intervals, including the endpoints: $10\text{--}100 M_{\odot}$ (light), $10^3\text{--}10^4 M_{\odot}$ (intermediate), $10^4\text{--}10^6 M_{\odot}$ (heavy), and $10^2\text{--}10^6 M_{\odot}$ (broad). The grid uses $z_{\text{seed}} = 30$, spins $a = -1, 0, +1$, merger multipliers $B_{\text{merge}} = 1, 2$, and average Eddington ratios $0.3, 1$, and 2 , with the efficiency prescription in Section C.1.

The intervals overlap and carry no probability weights; the broad interval covers several possible seed-formation channels. A required seed below an interval’s lower bound is incompatible: even its smallest seed would grow beyond the observed mass under those assumptions. We report the percentage of objects whose required seeds fall within each interval, using central mass estimates without propagating their errors. These percentages describe the literature sample. Population fractions would require the contributing studies’ selection and detection probabilities.

The two samples show similar patterns (Fig. 6). For heavy seeds at $a = +1$ and $f_{\text{Edd}} = 1$, changing B_{merge} from 1 to 2 raises compatibility from 58 to 71 per cent in the primary sample and from 59 to 71 per cent in the exploratory sample. Higher accretion rates can reduce compatibility when even the smallest seed grows beyond the observed mass.

Compatibility across growth assumptions

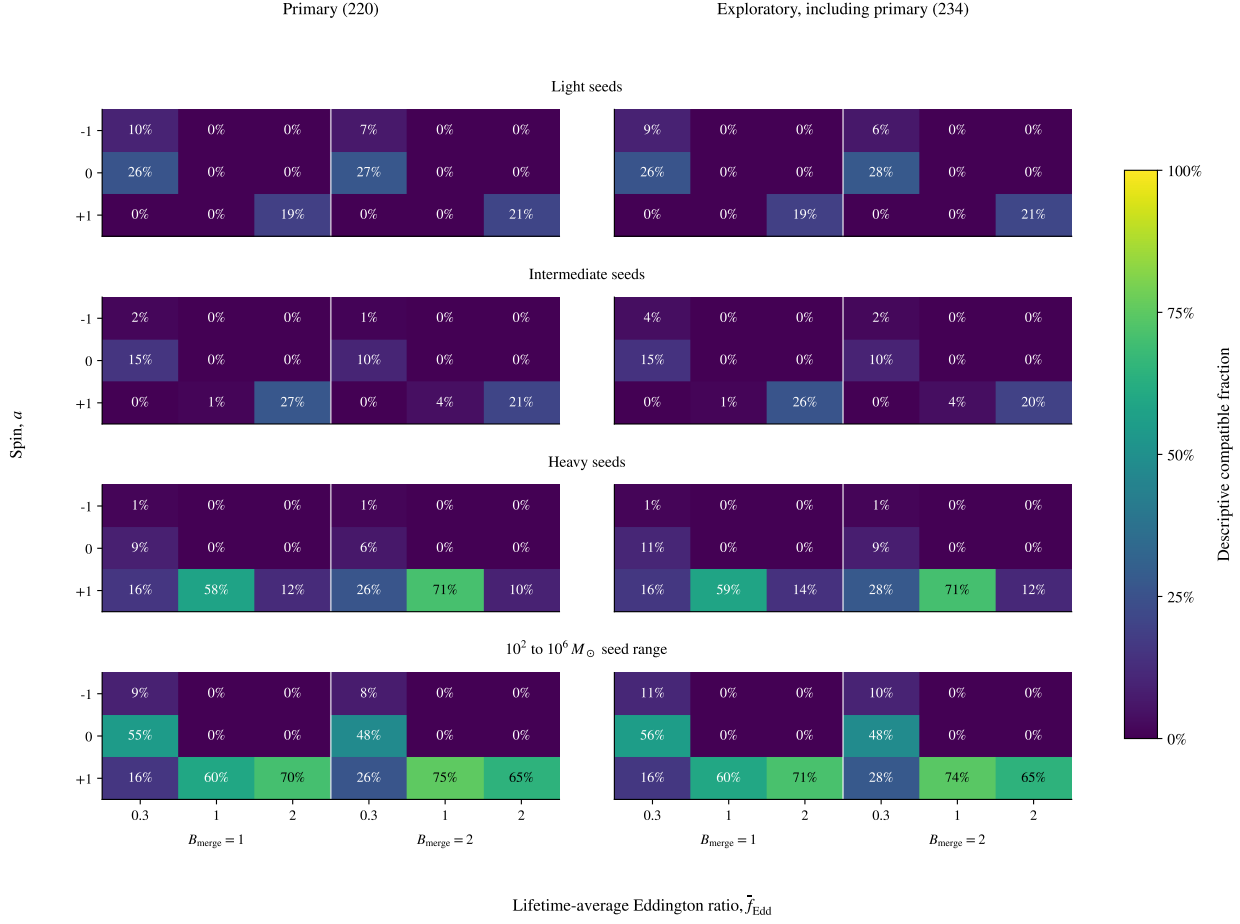


Figure 6: Constant-state compatibility fractions in the primary sample (left) and the inclusive exploratory sample (right). The exploratory sample includes all primary objects. Each column specifies merger multiplier B_{merge} and fixed luminosity ratio f_{Edd} ; rows specify ideal spin-dependent efficiency, with the supercritical prescription in Section C.1. Seed intervals overlap and have no assigned probability weights.

Appendix D Seed mass and starting redshift

Figure 7 extends the two starting redshifts in Figure 3 to $11 \leq z_{\text{seed}} \leq 30$ for UNCOVER-20466 and GN-z11. Later growth onset increases the required average Eddington ratio; larger seeds reduce it. Maps for all objects are available in the online galleries.

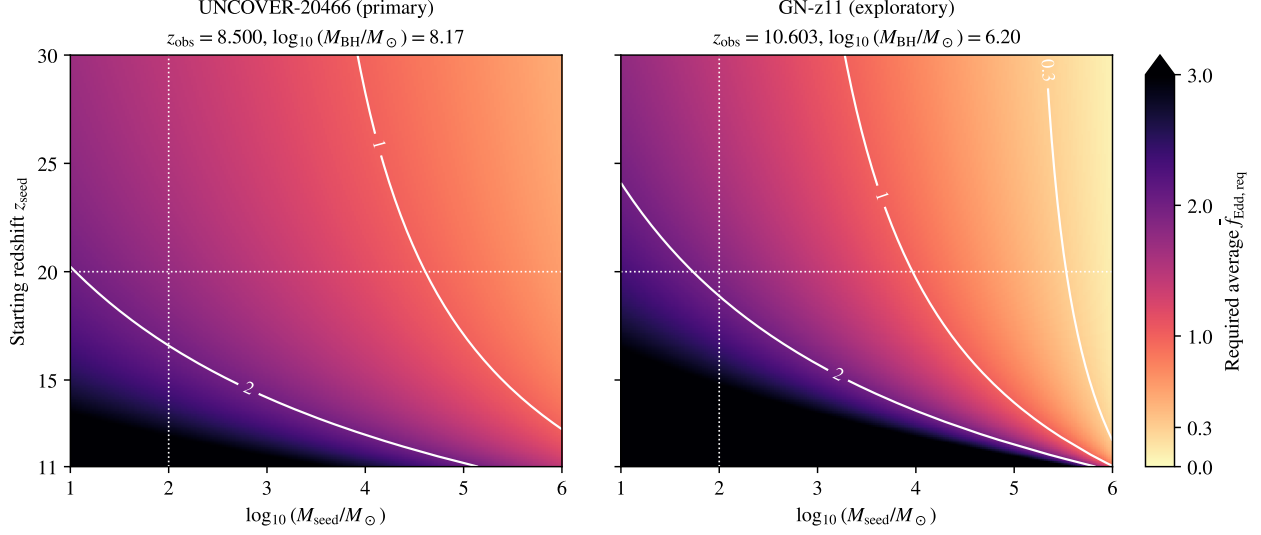


Figure 7: Required average Eddington ratio for UNCOVER-20466 (primary sample) and GN-z11 (exploratory sample), using their adopted central masses and redshifts shown above each panel. Both panels use $\epsilon = 0.1$, no merger boost, and the same axes and colour scale. White contours mark required ratios of 0.3, 1, and 2; dotted guides mark a $100 M_{\odot}$ seed and $z_{\text{seed}} = 20$. Colours saturate above a required ratio of 3. Mass uncertainties are not propagated in these maps.

Appendix E Data and code availability

The catalogue and code are available at <https://github.com/nniickels/highz-accretion-atlas>. Catalogue data and source records are in `data/`; manuscript results and figures are in `paper/analysis/` and `paper/figures/`. The analysis code is in `src/`, with workflow notebooks in `scripts/`.

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